MKGM31129 · ESSEC MiM 1 · FALL 2026

Marketing Management

Complete revision notes · Arnaud De Bruyn

ASSESSMENT Case study 1 : 20 % · Case study 2 : 20 % · Term project : 30 % · Quiz en fin de séance : 30 %. Pas d'examen final. Notes individuelles modulées par les peer evaluations.

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01 S1 · The course, marketing and response models

What marketing means in this course

Arnaud De Bruyn defines marketing as the art of **identifying (or creating), acquiring and keeping customers, profitably**. All four verbs matter: identify (understand the market), create (generate demand), acquire (conquest), keep (retention). The word "profitably" rules out marketing that wins customers at a loss.

Under this definition, marketing covers: segmentation, positioning, brand management, advertising and campaign effectiveness, sales and targeting, customer relationship management (CRM), channel management, new product development, pricing and value management.

Why the stakes are huge

Indicator	Order of magnitude	To picture it
Global marketing investment (media, martech, salaries, agencies)	$3.7 to 3.9 trillion per year	≈ $10.6 billion per day, the GDP of France
Paid media alone	$1.1 to 1.2 trillion per year	More than the combined revenue of Alphabet, Nvidia and Amazon

At these amounts, allocating a few percent better is worth enormous sums: that is the case for models.

Marketing, strategic marketing, data-driven marketing

- **Strategic marketing**: a long-term, forward-looking approach that plans and allocates marketing resources.
- **Data-driven marketing**: vision, intuition and creativity remain essential, but they are no longer enough on their own. Managers are now expected to back their choices with evidence.
- Some solutions replace human judgement (programmatic segmentation, real-time bidding, recommender systems, generative AI built into production). For strategic decisions (segmentation, product design, pricing, allocation, targeting), analytics **informs and sharpens** the manager's judgement without replacing it.

How managers decide

Decisions rely on **experience**, **individual mental models** (self, colleagues) and **collective practices** (shared rules, ratios). These shortcuts produce **biases**.

The pro-rata rule. A common rule allocates the promotion budget in proportion to market share: region A (30% share) → 43% of the budget; region B (40%) → 57%. But a region with 0% market share gets nothing, even if it offers the highest growth potential. The rule ignores the **marginal response** of sales to spending.

Every decision has consequences and **opportunity costs**: choosing A means giving up the consequences of B and C. A **response model** lets you compare consequences before acting.

Response models

A **model** is a stylised representation of reality, easier to manipulate and explore than reality itself, built for a specific purpose. A **response model** links a marketing effort (budget, price, number of sales calls, etc.) to an outcome (sales). Typical shape: an S-curve between a minimum sales level (no effort) and a maximum (saturation), passing through the current point.

Case	Problem	Approach	Result
Heinz	Is US promotional spending effective?	Response analysis and resource allocation	Spending –40% **and** market share from 34% to 37%
Courtyard by Marriott	Running out of good locations for full-service hotels	Conjoint analysis to design the hotel	Fastest-growing mid-range chain, over $1bn in sales, +4 pts of market share, 5 copycat chains
Charity	Soliciting is expensive and donors watch how gifts are used	Choice model to solicit only profitable donors	Solicitations –40%, donations +9% in year 1
Represent (luxury streetwear)	Which channel really pays?	Marketing mix modelling (MMM)	Meta twice as profitable as estimated, +44% of sales attributed to marketing, +20% ROI

Three sentences to remember. Models are not meant to be trusted, they are meant to be used. All models are wrong, but some are useful. Models don't make decisions, managers do: **50% data, 50% manager**.

Course organisation

Session	Date	Content	Deliverable
S1	5 Oct.	Introduction, Allegro exercise	Quiz
S2	12 Oct.	Segmentation	Term project groups before the session; quiz
S3	19 Oct.	Conjoint analysis, in-class survey	Survey (case 1)
S4	26 Oct.	Case 1 debrief, positioning maps	Case 1 due 25/10; quiz
S5	2 Nov.	Pricing, term project coaching, survey	Pitch due 5/11
S6	9 Nov.	Scoring and targeting	Quiz
S7	16 Nov.	Case 2 debrief, media planning, MMM	Case 2 due 15/11; quiz
S8	23 Nov.	The dark side of marketing	Quiz
S9–S10	30 Nov., 7 Dec.	Term project presentations	Project due the Sunday before

Tools: **Enginius** software (licence provided by ESSEC: create an account, validate it, link the licence), reference textbook (no need to buy it), "Virtual TA" for the term project. Electronic devices banned unless stated otherwise.

02 S2 · Needs-based segmentation

The managerial problem

Two questions: how do we understand the **diversity of customers' needs and characteristics** and divide a market (segmentation)? How do we identify the best customers and the most relevant segments for an offer (targeting)?

A company deals every day with hundreds, thousands or even millions of customers. It cannot treat them all **the same way** (different needs, profiles and potential; the "one size fits all" era is over) nor all **differently** (too many; rarely enough resources to personalise everything: product, offer, price, communication).

Definition. Segmentation is the process of classifying customers into groups (segments, clusters) so that customers in the same segment are **similar** enough to be treated the same way (economies of scale, consistency), and **different** enough from those in other segments to deserve differentiation.

The five qualities of a good segment

Quality	Question to ask
Distinct	If two segments have similar customers, why treat them separately?
Homogeneous	Otherwise its members cannot be treated the same way, which defeats the purpose.
Identifiable	A segment is not just conceptual; if its members cannot be found or targeted, it is worth little.
Substantial	Not necessarily in number of customers: at Melijoe, 3% of customers make 38% of sales.
Useful, actionable, insightful	Too many segments kills segmentation. What's in it? What do I learn? How does it help my strategy?

There is **no optimal segmentation**. A deck of cards can be segmented by suit, by value, face/non-face cards, red/black: each split is right for a given game. A segmentation is good **for a purpose**.

What segmentation is for

Short term	Long term
Resource allocation, sales force allocation, call planning, direct-marketing targeting, channel assignment, communication programmes, pricing	Strategic positioning, entering a new market with few resources, product development, new markets and niches, meeting needs differently, spotting emerging needs (a small group of current customers can signal the market's future needs)

The 7-step process (STP)

1. **Define the segmentation problem**: the critical step. Internal assessment: objectives, resources, constraints. Data review: what primary and secondary data are available?
2. **Identify the data needed**: socio-demographic or firmographic? Past behaviour, purchase data, checkout data, online traces, usage data (IoT)? Deep needs, benefits sought, perceptions?
   If the data **already exist** → *database marketing* (e.g. spot heavy buyers or deal-seekers in the customer database; mostly behavioural data). Otherwise → *market research* on a small random sample, then extend the learnings to the whole market (e.g. understand how my brand is perceived across segments).
3. **Run the study** if needed: qualitative first (interviews, identifying deep needs, the decision process, choosing criteria and questions), then quantitative (sampling plan, questionnaire, data collection).
4. **Build the segmentation dataset** with two types of variables (see box).
5. **Define** the segments: how many? on what bases?
6. **Describe** the segments: how do we recognise and reach them?

7. **Implement**: targeting, then positioning.

Bases and descriptors, the key distinction.

- **Segmentation variables (bases)**: say **why** segments differ. Needs, wants, benefits sought, solutions to problems, usage situation, usage rate, decision process, past behaviour.
- **Discriminant variables (descriptors)**: say **how to find, reach and identify** the segments. B2C: age, income, education, occupation, lifestyle, media habits, usage occasions. B2B: industry, size, location, organisational structure.
- Segments are formed **on the bases only**. Descriptors do not describe what customers want or why they buy.

Example sentence: "Segment X values prestige and peace of mind and is not price-sensitive (bases). It is mostly made up of high-income men who have been loyal customers for over 3 years (descriptors)."

The B2B catalogue example

Context: IT sector, assembly parts, direct sales through a catalogue. Two questions on a 1–7 scale: (1) how many items do you want to find in our catalogue (1 = few but easy to find, 7 = very wide choice); (2) how much technical support do you expect (1 = none, 7 = a lot). Each respondent is a row of the table and a point on the chart. With 8 respondents, we intuitively see 3 segments:

Segment	Profile	Suggested name
1	Small catalogue, lots of support	"Clueless?"
2	Large catalogue, lots of support	"Demanding?"
3	Large catalogue, little support	"Self-service?"

Hierarchical clustering, step by step

1. At the start, each customer is a segment (8 segments).
2. Merge the two segments whose merger causes the **smallest loss of information** (the two closest).
3. Repeat until a merger would cause an unacceptable loss of information.

At the extreme, everything can be merged into one segment (useless): segmentation is a **trade-off** between mass marketing (1 segment) and one-to-one marketing (1 customer = 1 segment).

In practice there are many more questions (dimensions) and respondents (points): we need a systematic method and formal definitions of similarity and information loss.

```
Similarity = Euclidean distance between two respondents a and b:
d(a,b) = √[ (x_a - x_b)² + (y_a - y_b)² + … ]
The larger the distance, the lower the similarity.
The formula accepts as many questions as needed.
```

- Consequence 1: questions must use the **same scale**, otherwise they must be **standardised**.
- Consequence 2: an irrelevant question weighs as much as a relevant one. Choose your questions carefully.

Calculation. Respondent A = (1, 7), respondent B = (4, 3). $d(A,B) = \sqrt{[(1-4)^2 + (7-3)^2]} = \sqrt{(9 + 16)} = \sqrt{25} = \mathbf{5}$.

Reading the outputs

- **Dendrogram**: a tree read from bottom to top (each observation alone at the bottom, a single segment at the top). The height of each merger is the distance between the two groups merged. A **big jump** = merging two very distant groups = large information loss → stop **before** it (the "sudden jump" method).

- **Scree plot**: for each number of segments, the sum of squared errors (SSE) between each observation and its segment's centre. Look for an **elbow**, beyond which adding segments no longer reduces heterogeneity much. There is not always an elbow.
- The ideal number of segments combines **statistical fit** (what the data say), **managerial relevance** (what makes sense) and **targetability**. When they disagree, it is a judgement call.
- **Summarise** each segment by the average of its members (centroid); Enginius colours in green (higher) or red (lower) the variables that differ significantly from the rest of the population.
- **Name** segments carefully: names stick.
- **Segment map**: first two dimensions of a PCA on the segmentation data (in the example, 78.3% of variance explained); useful to see correlated variables, misleading for choosing the number of segments.

Describing and finding segments

- Segments are described by their **descriptors** (who they are, how to reach them) and by **personas**: a fictional character representing a type of user.
- Problem: how do we assign to the right segment all the customers who did not answer the survey? Bases are known only for respondents; descriptors are often available for everyone (files, secondary sources).
- **Discriminant analysis**: predicts segment membership from descriptors (in Enginius, a multinomial logit with the largest segment as reference). Quality: **confusion matrix** (actual segment × predicted segment) and **hit rate** (share correctly classified).

Enginius output from class. 40 respondents, 6 segmentation variables (choice, electronics, furniture, service quality, low prices, return policy), 3 descriptors (professional, income, age). Segments: 18 (45%), 8 (20%), 14 (35%). Segment 2: very attached to service quality (8.50); segment 3: low prices (8.29) and return policy (6.29), younger (30.9 years) and less wealthy (32,143). Confusion matrix: 13 + 4 + 12 = **29 correctly classified out of 40 → hit rate 72%** (72%, 50%, 86% by segment). Segment 2 is the hardest to spot from the descriptors.

The Acxiom case and the political campaign

1. **Deep-needs survey** in the US: 3,000 respondents, 140 questions (view of the future, definition of success, values, priorities, past vote).
2. **Segmentation**: 5 segments ("Extending opportunities to others" 37%, "Working within the community", "Achieving independence", "Focusing on family", "Defending righteousness" 16%), then 10 "tribes".
3. **Swing voters**: no point convincing those already won (Democrats for the 1st segment, Republicans for the last); target the three middle segments.
4. **Discriminant analysis**: buying descriptors (ChoicePoint: taxes, court rulings, civil records; Acxiom: purchases and lifestyles of 200 million Americans). More likely Democrat with a cat, sushi, Apple; Republican with a dog, a Pontiac, a gun, church.
5. **Targeting**: buy the addresses of 20,000 people likely to be "Working within the community" in a close district for targeted mail; buy ad spots during the shows watched by "Focusing on family".

Classic mistakes. Forming segments on age or income. Confusing segmentation (forming groups) with targeting (choosing whom to serve). Keeping too many segments. Forgetting to standardise different scales. Including off-topic questions that blur distances. Naming segments carelessly.

03 S3 · Conjoint analysis

The problem

Imagine you are designing an app: which features should it include, at what price? How do you extract market preferences through research? Managerial questions:

- Understand and **quantify** what consumers value, like and dislike.
- Link **preferences** to **choices**.
- Estimate the potential **market shares** of new offers and **cannibalisation**.
- Identify new product opportunities.

Use in business: 85% of managers consider conjoint analysis the best method at the product strategy stage; 60% of new product projects use it; 90% of managers find it reliable.

Products, attributes, levels

A **product** is described by **attributes**, each with several **levels**. Examples: colour (red, blue, green, black); warranty (none, 1 year, 3 years); brand (Samsung, Acer, HP); stopovers (direct flight, 1 stop, 2 or more); cooking (blue, rare, medium, well done); customer rating (5 stars, 4 stars...).

The pizza. Size (small, large) × topping (four seasons, pepperoni, mushrooms) × price (€9, €12, €15). A product is just a **bundle of levels**: "small pepperoni at €9", "large mushroom at €15"... There are 2 × 3 × 3 = **18 possible products, and nothing else**.

Partworths

Each level is given a **partworth** (partial utility). A product's utility is the **sum** of the partworths of its levels.

Attribute	Level → partworth
Size	Small 0 · Large 10
Topping	Four seasons 30 · Pepperoni 40 · Mushrooms 0
Price	€9 50 · €12 30 · €15 0

Large (10) + pepperoni (40) + €9 (50) = **100** · Small (0) + four seasons (30) + €12 (30) = **60** · Small (0) + mushrooms (0) + €15 (0) = **0**. The first product is "more likely" to be bought.

- By convention, preferences are scaled from 0 to 100: worst levels = 0, sum of the best levels = 100 (done by the software).
- Obviously, people prefer to pay less. What matters is the **strength** of that preference compared with the other attributes.
- **Attribute importance** = the gap between its best and worst partworth. Here: price 50, topping 40, size 10. An attribute whose levels all have similar partworths matters little, whatever the respondent says.
- Preferences vary from customer to customer; customers can be segmented on their partworths.

Estimating partworths

Asking directly "how important is price to you?" is **not reliable**: everyone says everything matters, since they don't have to trade off. Partworths are estimated **indirectly** ("considered jointly").

1. **Create profiles** (bundles): let the software choose the list, to ensure the conditions for the regression (enough profiles, no multicollinearity).
2. **Have the profiles rated** on a scale (0–100, 1–10): these are the preference scores.

3. **Regression**: rating = constant + $\Sigma\ \beta_j \times X_j$, where $X_j = 1$ if level j is in the profile, 0 otherwise. Each β is the level's contribution to the rating: that is the partworth.

```
Minimum number of profiles to rate ≥ number of parameters to estimate
Parameters = Σ (number of levels of each attribute - 1) + 1 (constant)
Pizza: (2-1) + (3-1) + (3-1) + 1 = 6
5 attributes × 5 levels: 5 × 4 + 1 = 21
```

Once the β are estimated, we can predict the appeal of **any profile**, even one not rated or that does not exist yet. Variant: **choice-based conjoint** shows sets of profiles and asks which one you would choose; it estimates partworths with a **binary** regression on choices instead of a **linear** regression on ratings.

From preferences to choices

Rule	Principle	Context
First choice	Each customer picks their favourite product (100% for it)	High-stakes decision: house, car
Share of preference	Probability proportional to utility	Strong variety-seeking: meals
Logit	Probability from a logistic function of utility	In between: clothing
Alpha-rule	Calibrated on current market shares	More accurate and versatile, if shares are known

Share-of-preference calculation. A customer has utilities of 60 (A), 30 (B), 10 (C). Probabilities: 60%, 30%, 10%. Under first choice: 100% for A.

Simulations

Once attributes are defined, data collected, and preferences estimated and linked to choices, we can simulate market shares if we raise a price, change a product, launch a new product, or if a competitor enters, and measure cannibalisation.

- **Example 1**: people prefer flying to taking the bus; we turn trip no. 3 into a flight at a slightly higher price and measure the share gained.
- **Example 2**: a low-cost newcomer arrives. If the 3rd product (mountains, €450, by bus) belongs to the same company as the new one, cannibalisation is strong.

Running a conjoint study in 3 steps

1. **Design**: choose relevant attributes (customers, managers, experts, secondary sources), **4 to 6 attributes maximum**; choose levels, **2 to 4 per attribute**, realistic, covering the market's options, ideally the same number per attribute; let the software generate the profiles; if some profiles make no sense, revise the design.
2. **Collect and estimate**: questionnaire, sample of target customers, regression.
3. **Use**: choice rule, reproduce the existing market, scenarios (new product, new entrant, changes), segmentation on preferences. An offer with a small overall share can **dominate a niche** (for example people who hate flying).

Classic mistakes. Correlated attributes ("container size" and "amount of liquid"): some combinations make no sense; only price may be correlated. Forgetting a key attribute or the competitors. Too many attributes or levels. Thinking that "preferring to pay less" is informative: only relative strength matters.

04 Case study 1 · Internship preferences (due 25/10)

The first case applies **conjoint and segmentation** to students' internship preferences. The twist: you are both the analyst and the data, since the class designed and filled in the survey in S3. The internship market is **two-sided**: students compete for the best positions, but employers also compete for talent by adjusting attributes (pay, location, flexibility, mentoring, prospects).

Question 1 · Trend analysis

Before the data: what is happening in the internship market? Recent, credible sources (business press, industry reports, employer surveys). What do employers offer, and how have expectations changed? Focus on trends linked to the survey's attributes. A good analyst never interprets preferences in a vacuum.

Question 2 · Estimate the partworths

- When the results are imported, Enginius creates a ready-to-run "Conjoint analysis" model with the design (attributes, levels) and the responses.
- Read three outputs: "Average preference partworths" (average partworth of each level), "**Attribute importance**" (relative importance of each attribute), "Detailed preference partworths" (individual partworths, excerpt on screen; full table in the Excel export, needed for question 5).
- Partworths are scaled at the individual level: directly comparable across respondents.
- Deliverable: on average, what do students care about most? Support it with both charts.

Question 3 · Simulate competing offers

1. Create a data block: one row per offer, one column per attribute; headers **identical** to the attribute names; each cell = an existing level, spelled the same way (you cannot simulate a level that is not in the survey). Tip: run the conjoint model generator with "Prepare simulation placeholders" to see the format.
2. Invent **4 to 6 offers**, realistic and varied.
3. In the model: tick "Run simulations", choose the block under "Existing product profiles", type "Existing products only", rule "**First-choice**" (clear-cut results).
4. Read "Simulations with existing products": predicted share of each offer = share of students it would attract.

Question 4 · Optimal offer under a budget constraint

1. Copy the "conjoint design" block, rename it ("Restricted levels"), delete all salary levels except the lowest; keep at least one level per attribute and leave the other attributes untouched.
2. Simulation type "...With optimal products", tick "Restrict usable levels" and choose this block.
3. Read "Simulations with optimal products": the best profiles found and how they perform against the existing offers.

Competitive logic. "The best possible internship" ≠ "the best possible internship given the offers already on the market". If every competitor already offers flexible hours, offering flexibility, even if highly valued, creates no advantage: it is the standard. A product serving a small but uncontested segment can be worth more than one that appeals to a large, highly competitive segment.

Question 5 · Segment on preferences

- Averages mislead: a moderately important attribute may matter moderately to everyone, or be the result of two groups with opposite preferences. "The average student does not exist."
- **Bases** = individual partworths (two students are in the same segment if they make the same trade-offs). **Descriptors** = gender, age, nationality, admission track, motivations, goals.
- Prepare two blocks: partworths (Excel export, one row per respondent, ID in the 1st column) and descriptors (same IDs, same order, same number of rows; Enginius refuses otherwise).

- "Segmentation" model: **do not standardise** (partworths are already on a common scale; standardising would distort the attributes' relative importance). Tick "Descriptor data".

Recommended method for the number of segments. Run with 2 segments and understand them. Move to 3: one segment stays intact, the other splits (visible on the dendrogram). Does the split make sense, is it managerially useful, does it reveal hidden heterogeneity? If yes, keep 3 and try 4; if not, stop. Then name each segment and give it an image, a short description and a typical profile you could present to managers.

Expected conclusion

Step back: what would these analyses tell an employer designing its internship programme? And you, as a candidate: where do your partworths place you, and does your segment look like you?

05 S4 · Positioning and perceptual maps

Managerial questions

- How do customers form their perceptions?
- How do competitors position themselves relative to these perceptions?
- Are there opportunities for a new offer, a better positioning or a repositioning?
- How do we link **perceptions and preferences** to explain behaviour?

The positioning map

Starting example: a table of average perception scores for beer brands in Spain. Unreadable as it is; a map makes it obvious.

- "Competitors" (products, companies, offers) are placed on a map along dimensions **relevant to customers**.
- **Short distance = fierce competition.**
- A simple tool with powerful insights: easy to read, graphical, 2D; humans understand charts better than numbers; easy to communicate, defend and act on (the "aha!" effect). Gartner's "Magic Quadrant" is the best-known example.
- Limits: are 2 dimensions enough? What data should be collected? How do we link perceptions, preferences and market shares?

Why perceptions matter

Case	What it shows
Volvo	Positioned on safety forever. Yet GM ran the first crash test (1934), Saab built the first safety cage (1949), Ford had the first airbag-equipped fleet (1971). Volvo = safety in customers' minds.
Drilling company	Breakthrough technology, faster and cheaper. It saw itself as dominant, but to customers the positioning was **not credible**. The product became a big success **once the price was raised**.
Débitel	Low-cost reseller with no network of its own, buying airtime from SFR (no. 1 for quality). Perceived network quality: SFR 8.0/10, Débitel 5.9/10, on the **same network**. Bought by SFR, then La Poste Mobile.
Cat litter	White grains and blue grains: plain clay painted blue, with no chemical property or scent, yet perceived differently.

The principles. We live in an **attention economy**. You cannot be everything to everyone: focus, clarity, brevity, consistency. **Truth matters less than perceptions.** Adapt to customers' perceptions: "**wishful marketing**" (communicating what you would like to be rather than what you are) is a death wish; trying to reshape the market's perceptions is an expensive death wish. Since attention is short, the message must be focused and perceptions prevail: it is **critical to map customers' minds**.

Correspondence analysis

The human brain is poor at visualising numbers. Correspondence analysis (also called perceptual mapping, MDS, factor analysis or PCA depending on the variant) **reduces a large number of variables** (attributes, perceptions) to a few key dimensions by analysing their correlations.

- **Brands** are points. Close brands = similar perceptions = competing on the same dimensions.
- **Attributes** are vectors (bitter, refreshing, nightclub, price…) showing the positioning dimensions.
- Vectors pointing in the **same direction**: strongly correlated (expensive, nightclub, international). **Opposite directions**: negatively correlated (bitter vs young). **Perpendicular**: uncorrelated. Intermediate angle: somewhat correlated.
- The **further from the origin** a point or vector is, the more significant it is.

- To compare brands on one dimension: extend the vector and draw **orthogonal projections** of each brand onto that axis.
- We also try to give meaning to the horizontal and vertical axes (e.g. service quality vs product quality).

Two properties to understand (case 2). The map is **relative**: a brand's position only makes sense relative to the other brands in the set; changing the set of brands changes the map. The dimensions are **not given but interpreted**: the analysis returns unlabelled axes; naming them is the most consequential analytical judgement.

Linking perceptions and preferences

Some dimensions correlate with preferences, choices and market shares, others do not. Coronita is well positioned on "international", but nothing guarantees the target population values that feature. We add **preference data** (one row per respondent). Each preference row shows what one customer wants; some dimensions don't matter; preferences are **heterogeneous**.

A position can be **distinctive and worthless**: unique, but wanted by nobody. What matters: being differentiated **and** aligned with the direction of preference, and knowing whether someone else is already there. An attractive but crowded region can be worth less than a slightly less attractive but completely empty one.

Running a study in 3 steps

1. **Design**: brands the segment knows and can evaluate, without forgetting the leaders; relevant descriptive attributes, including the attributes most representative of the brands (with Volvo, include "safety"); attributes tied to the perceptions on which products are usually judged; scale (e.g. 1–7).
2. **Collect**: use the software to create the questionnaire template; perceptual data (one matrix per respondent, averaged over the sample) and preference data (one row per respondent).
3. **Analyse**: how many dimensions (if 2 are not enough, move to 3; look at the **variance explained**); interpret brands and competition (who is close to whom?); interpret the dimensions (correlated, opposite, perpendicular? meaning of the axes?); spot poorly explained brands or dimensions (bad positioning, unknown brand, missing or wrong data).

Tips from the coaching deck (term project)

- Do not include brands that are not in the same perceptual space (Fiat and Ferrari: a pointless map).
- Stick to a single, coherent experience: for airlines, either the flight experience or loyalty programmes, not both. For car brands, measure brand perceptions ("reliable") rather than model features ("spacious").
- Better **a few knowledgeable respondents** than many uninformed ones: otherwise everything lines up on a single good/bad axis.
- Exclude options unknown to the respondent, include a preference question, add descriptors.

06 S5 · Pricing

Price and the "value salami"

Price is the most important element of the marketing mix: it is the **only** one that appears in **every component** of the profit function.

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Profit = (Price - Cost) × Volume
Price acts on the unit margin (Price - Cost), on volume (demand),
and, through volume, on cost (economies of scale).
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- **Minimum selling price** = the seller's reservation price; **maximum buying price** = the buyer's reservation price. In between: the **value created**, to be shared.
- Price determines the share captured by the seller and the share captured by the buyer. **No value creation, no sale.**
- We must also distinguish **perceived value** from **true value** (perception gap), and production and distribution costs.

To capture more value…	Condition
Cut costs	Without cutting value, and without passing all the savings to the customer
Raise the price	—
Increase value	And capture part of it
Increase perceived value	Narrow the perception gap, enlarge the perceived share

Pricing strategies

Strategy	Definition	Key points
Cost-based	Cost + mark-up (% or fixed amount), "cost plus"	Very common (retail), but ignores price elasticity of demand: it only leads to maximum profit by chance.
Competition-based	Price set on competitors' prices (market price, imitation)	Typical of commodities (oil, water). Frozen market: raise your price and you lose a lot of share; cut it and the others follow. Price-war spiral ("a war nobody wins; even if you win, you lose") or stagnation. Remedy: **differentiation** (style, quality, features, brand) reduces price sensitivity; the price premium customers accept measures brand equity.
Demand-based	High price when demand is strong, low when it is weak	Consequence: **if no customer complains about your prices, your price is sub-optimal** (some potential customers should find it too expensive).
Value-based	Price based on the value a given customer places on the product	Mainly B2B, when the seller has pricing latitude. Measured with conjoint or **value-in-use**. Communicating and demonstrating value also narrows the perception gap.
Price discrimination	Different prices for customers or segments with different reservation prices	The "holy grail" of demand-based pricing.

Value-in-use: the O-rings. A chemical plant uses 200 seals at $5, replaced at every maintenance shutdown every two months; each shutdown costs $5,000. A new seal lasts twice as long (replaced every 4 months). Over 4 months, the old system costs 2 × (200 × 5 + 5,000) = **$12,000**. The new one involves a single shutdown ($5,000); that leaves 12,000 − 5,000 = $7,000 for 200 seals: **value-in-use = $35 per seal**, not $10 as competition-based pricing would suggest ("twice as good, twice the price").

Discrimination: the ice-cream seller. 11 customers with reservation prices of €10, 9, 8, … 1, 0. Cost: €2 per ice cream. Optimal single price €6 (5 sales × €4 = €20) or €7 (4 × €5 = €20): profit **€20**. If segment 1 (€10 to €6) can be told apart from segment 2 (€5 to €0): €6–7 for the first (€20), €4 for the second (sales to the €5 and €4 customers: 2 × €2 = €4) → **€24 (+20%)**.

- Done well, discrimination lets you charge some customers less without cutting the margin on those who pay more: less money "left on the table". Perfect discrimination is **Pareto-efficient**.
- **Limits**: reservation prices are hard to identify; observable characteristics correlate poorly with reservation prices; hard to target a segment with posted prices; you must prevent **arbitrage** (low payers reselling to others); illegal if based on sex or ethnic origin; feeling of unfairness.
- **Types**: geographic (if arbitrage is hard: barriers, transport); time-based (non-storable goods: hourly, daily, seasonal prices, depending on the purchase cycle); by customer group (age, income, student status); individual.

Pricing tools

Tool	Gives	How it works	Strengths	Limits
Conjoint analysis	Preferences for different prices and attributes	The price partworth captures willingness to pay	"What-if" scenarios, includes competition; the most powerful tool	Heavy to design
Gabor-Granger	Optimal price	Purchase intention (1 = never … 4 = probably, 5 = definitely) at several prices, shown **in a random order** that differs for each respondent (reduces rationalisation). Converted into probabilities (e.g. 4 = 20%, 5 = 50%, otherwise 0%), ideally calibrated on real data. → demand curve, elasticity, revenue, profit (with costs).	Simple, one measure per price, robust, proven	No margin of error (it can be added); assumes accurate data; you can **interpolate but not extrapolate**: if the optimum lies above the highest price tested, nothing can be said
Van Westendorp	Psychological price and acceptable range	Four open questions: at what price is this product **too cheap** (not credible), **cheap**, **expensive**, **too expensive** (unaffordable)? Plot the cumulative curves and read the intersections.	No imposed range, no anchoring; not very sensitive to extreme values; a good framework when better data are lacking	No econometric data; what if respondents have no idea? Best seen as an extra benchmark

Reading Van Westendorp. Optimal psychological price = intersection of "too cheap" and "too expensive": the point where opinions balance out. **Acceptable** range = between the two outer intersections: "cheap × too expensive" and "too cheap × expensive". (The PMC/PME points: price marginally cheap / expensive.)

Gabor-Granger, calculation. At €40, 30% purchase probability; at €50, 22%; at €60, 15%. Revenue per potential customer: €12, €11, €9. Without costs, €40 maximises revenue among the prices tested. With a unit cost of €20: expected margins €6, €6.6, €6 → €50 maximises profit.

Tips for pricing in the term project

- Define the product clearly ("text with image" block).
- Choose a **fixed-price** context: in B2B, where prices vary with the client's size or industry, an "optimal price" makes no sense. Works best in B2C, standardised products.
- The highest price tested must be **high enough**: you interpolate, you don't extrapolate.
- Segmentation and pricing can be combined: build the segments, export, then run a separate price study for each segment.

07 Case study 2 · Everyday Sneakers (due 15/11)

Second case: **positioning and pricing** in the everyday sneakers market. The class rated the main brands on attributes, said which ones they would wear, and gave their willingness to pay for **a single pair** from the brand of interest (announced in class).

Two levels of analysis. Positioning concerns a **competitive set**: it only makes sense if several brands are mapped together. **Pricing** concerns **one defined product at a posted price**: the data exist only for the brand of interest's flagship model.

The context: a converging market

- Sneakers are both **technical** products (measurable performance goals) and **fashion** products, bought mostly for occasions with no sport at all.
- Historically, two worlds: **performance** brands (running, training, court; sports channels; promises of cushioning, stability, lightness) and **lifestyle** brands (silhouettes; fashion channels; promises of style, heritage, cultural belonging).
- This divide is dissolving: brands born in running end up in fashion stores; court/street brands target the same place in the wardrobe. Running is growing, lifestyle is stagnating or declining; the leader is losing share to more fragmented challengers.
- The "everyday" occasion (the pair you put on every morning without thinking) is the category's biggest and most repeatable purchase, and that is where brands from both worlds collide. Perceptions built in one world do not automatically transfer to the other.

The five questions

Q	Expected	Tool
1	Trend analysis: who is winning, who is losing, on what bases competition takes place (press, reports, financial disclosures, consumer surveys)	Desk research
2	Perceptual map: how many dimensions to keep, what share of the variation they explain	Correspondence analysis
3	Project preferences ("which brands would you wear most days?"): in which direction preference increases, which dimensions drive it, regions that are attractive but empty, or occupied but unattractive	Preference mapping
4	Position of the brand of interest and recommendation: what it should own, what it should concede, concrete actions, and what you would **not** do	Map + preferences
5	Revenue-maximising price; is that price consistent with the positioning?	Gabor-Granger

A brand's strategic options

- Strengthen on a dimension where it is already credible;
- Try to progress on a dimension where it is weak;
- Attack a gap;
- Defend against a neighbour.

Each option implies a different allocation between product development, communication and distribution.

Credibility limits repositioning. Perceptions move slowly and are expensive to change. A recommendation that requires a brand to be seen as something it has never been amounts to spending a lot for little result.

Watch-outs for Gabor-Granger

- Price levels are set in class and cover a wider range than the model's usual price.
- Intentions → probabilities → share of buyers at each price → **expected revenue = share × price**.
- If revenue is still rising at the highest price tested, the analysis **has not found a maximum**.
- Check the price against the positioning: a high price on a brand seen as affordable, or the reverse, signals an inconsistency (either the model or the intuition is wrong; it is up to you to say which).

08 S6 · Scoring and targeting

Reminder: needs-based segmentation

Observe segmentation variables (needs, profiles, behaviours), group similar customers, obtain manageable, homogeneous and distinct segments, then describe and name them.

In-class example. Variables: will your first job be in marketing? in finance? in strategy consulting? do you plan to start a start-up within 5 years? are you analytically minded? do you like marketing analytics? are you close to graduating? do you like this course? do you like the instructor? Result with 4 segments: the "**Marketeers**" (45%), the "**Entrepreneurs**" (close to graduating), the "**Dark Navy Suits**" (not entrepreneurial, the youngest), the "**No-no-nothing**" (9%). Question: for a new "Data-Driven Marketing" elective, which segment should be targeted, with what message?

The limit of needs-based segmentation

It helps understand similarities between profiles, but does not always answer: will they buy my product? what drives their choice? which brand will they choose? Because **similarity does not necessarily predict choices**. Example: three segments of the premium car market (luxury/pleasure, status, peace of mind) are useful to understand motivations and tailor advertising, but which one will buy the new Mercedes?

Choice-based segmentation

It groups customers by their **response probability**: similarities matter only insofar as they predict choices. On the data side, everything becomes a predictor (segmentation) and a **response** variable is added; the model decides what matters.

CART (classification and regression tree)

1. Choose the predictors (past behaviour, demographics…).
2. Choose the response to predict: buy or not, donate or not (0/1); brand chosen (A, B, C); purchase amount.
3. Put all customers in one big parent node.
4. For each predictor, split the node into child nodes and measure whether they are more homogeneous (within) and more distinct (between).
5. Keep the best split, measured by an index (entropy, Gini, RMSE).
6. Repeat on each child node; stop when there is no more improvement or not enough data.

Course example. Predictors: profiles (analytical mind…), intended specialisation, current satisfaction (course, instructor). Response: likelihood of taking an advanced elective (1 to 5). Satisfaction with the instructor: poor predictor. Intention to specialise in finance: slightly better. Intention to specialise in **marketing**: excellent predictor. After several splits: 5 segments. Replacing "favourite course" with "favourite brand" gives a direct marketing application.

Limits of CART. Each split reduces the node's population: going deep requires a lot of data. Risk of **overfitting** (finding by chance a relationship that does not exist). Only a subset of predictors is kept; the others still have predictive value but are ignored. Within each node, the response remains heterogeneous.

Choice models

A **choice model** is a mathematical model that predicts the probability of an observed choice from predictors. Components: the observed choice (buy/don't buy in direct marketing; brand bought in consumer goods); the predictors (demographics, attitudes, perceptions, market conditions such as price or promotion, past behaviour); the link between the two. The model predicts each customer's purchase probability and reveals the **weight** of each predictor.

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Logit model: P(choice = 1) = 1 / (1 + e^-(β0 + β1·x1 + … + βk·xk))
The x are observed, the choice (0/1) is observed, the β are inferred for the best fit.
A positive, significant β: the higher x, the more likely the choice.
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Example: take the elective? 306 respondents, 1 choice, 9 predictors. Main drivers: interest in marketing, analytical mind, strategy consulting, closeness to graduation; interest in finance (negative effect). Quality: **80% correctly classified** (114 correct "yes" + 132 correct "no" out of 306). More false positives (42 predicted "yes" but actually "no") than false negatives (18). Is something missing? The **multinomial logit** extends the model to a choice among several options (brand, product).

Using a choice model for targeting

1. Calibrate the coefficients on a sample of customers (or the same customers in another period).
2. Apply the model to a wider list.
3. Predict each person's choice probability and rank from least to most likely.
4. Target according to the objective.

Objective	Examples	Whom to target	What is maximised
Trigger a purchase	Catalogue for an order, sales call, donation request letter	The **most likely** customers	Profitability
Influence behaviour	Coupon to encourage trial, change a perception	The **undecided** (switchers)	Marketing impact

Why? Very likely customers are loyal anyway: no need to send them a coupon. Very unlikely ones will never buy: don't waste resources on them.

Implication of the logit. The first derivative of the logit function gives the marginal impact of a change (a marketing action). It is highest when the **probability is close to 50%**.

Scores and score classes

- A **score** characterises a customer's most likely behaviour or potential. At its simplest, the output of a choice model (purchase probability); often a combination of models.
- Fundraising example: probability of donating (choice model) × predicted amount if they donate (regression) = **expected donation**. Two donors can have the same score with different behaviours (50% × €100 = 100% × €50).
- A **score class** groups customers whose score falls within a range (A, B, C, D). Use: churn prediction in banks.

The charity case (Christmas campaign).

Letters sent	301,500	Donations received	17,200
Response rate	5.7%	Total raised	€992,000
Average donation	€57.7	Mailing cost	€182,500
Net margin	€809,500	ROI	+443%
Fundraising ratio	18.4% (18 cents to raise €1)		

Scoring models: probability and amount of donation, based on recency, frequency, monetary value (RFM), past activity, demographics. Score = probability × amount; donors ranked by decreasing score. Result: **20% of donors = 76% of donations and 86% of net margin**. Options: raise **€10,000 more** with a ratio of 15.4%; or keep the same net margin with a ratio of 11.3%, an ROI almost doubled and **€80,000 saved**.

How many segments?

Approach	Few	More	Extremes
Needs-based	3–6: big picture, strategy	10–30: target groups, personalise campaigns	Market = 1 segment (mass) · 1 customer = 1 segment (one-to-one)
Choice-based	Choice-based segmentation (3–30 segments) → score classes → individual scores		Average market response

Score classes vs choice-based segmentation: both group by choice probability and are simple to use; classes are more complex (models updated regularly), more complete and accurate (all the data), but less intuitive (who is in class A, and why?). It is the classic trade-off of **accuracy/performance versus simplicity/usability**.

09 Term project · pitch, method, questionnaire

The principle

Like a case study, but you choose the business problem, design the survey and collect the data. Groups of 4–5 formed according to your preferences (sign-up before S2). Pitch before Thursday 5/11 at midnight; coaching in S5; presentations in S9 or S10; submission the Sunday before.

The pitch: one single slide

- **The context**: the company, start-up or situation (what is the topic?).
- **The problem** and the question (where is the tension?).
- **The method** you plan to use.

Method	When to use it
Segmentation	The most common: varied profiles and preferences of customers or prospects
Conjoint analysis	Quantify trade-offs between attributes (price, brand, quality, power); launch a product easily described by attributes; predict market shares
Positioning	Understand the market structure and how competitors are perceived; link preferences and perceptions
Pricing	Find the optimal price of a defined product or service

Two good past pitches. Joon (Air France's subsidiary for young travellers against low-cost carriers, cabin crew in sneakers, USB ports): positioning study → in December the group predicted there was no viable positioning; Joon was reabsorbed 18 months after launch. **dVine** (wine-by-the-glass capsule machine): segmentation + pricing → no viable segment (connoisseurs reject capsules, others won't pay for the machine); the company pivoted to restaurants and bars.

Finding a good idea

- **Where is the tension?** No tension, no case: a gap between where the company is and where it wants to go, or uncertainty about the future (new product). A company doing very well has no problem to solve.
- **Don't try to save the world**: a precise, solvable problem (a niche, a price), not "turn IBM around".
- **But be ambitious**: avoid everyday student topics (cafeteria, transport, phones); read the business press.
- **One method, two at most**: a simple approach perfectly executed is better.
- **Use your experience, passions and network**, without confidential information.
- **Be useful**: a personal project, friends' start-up, family business.
- **Think about respondents**: you need answers quickly (a topic on farm tractors needs respondents other than students).
- **Stay professional**: no "map of the worst professors".

Designing the questionnaire (coaching deck)

Rule	Why / example
Same format, same scales for all segmentation variables	A "To what extent do you agree…" grid simplifies the analysis
Focus on one aspect and dig deep	15 questions on one aspect beat 20 on unrelated topics (unreadable segments)
No branching	Instead of "Do you use Instagram? If yes, how often?": "How often do you use Instagram? (never … several times a day)"
Avoid binary questions and checkboxes	Prefer a 1–7 scale: "I am trying to lose weight", "I exercise regularly"…; turn a list of concerns into a grid from "not at all" to "major concern"
Clear, exhaustive, non-overlapping categories	"18–25", "25–35": where does a 25-year-old go? Add "Other", "None"
Unambiguous questions, no jargon	Words that mean different things to different people = worthless data
One question = one idea	"I get up early because productivity matters to me" mixes two things (insomnia? productive night owl?)

For conjoint: don't forget key attributes, remember to code the competitors, avoid correlated attributes (except price), add descriptors to segment the partworths. For everything: **go deep, not wide** and read the Enginius tutorials ("RTFM"). The deck also provides generative AI prompts to generate segmentation items (12 to 25 Likert-scale statements), descriptors (10 questions maximum, no sensitive questions), brands and perceptual dimensions.

10 Summary · which method for which problem

Manager's question	Method	Data	Key output
Who are my customers, how do they differ?	Needs-based segmentation	Bases (needs) + descriptors	Segments, profiles, personas, hit rate
What do they value, which offer should I launch?	Conjoint analysis	Ratings or choices of profiles	Partworths, importance, simulated shares
How am I perceived versus competitors?	Positioning (correspondence analysis)	Brand × attribute ratings + preferences	Perceptual map, preference vectors
At what price should I sell?	Gabor-Granger, Van Westendorp, conjoint	Intentions at several prices; price thresholds	Optimal price, acceptable range
Who will buy, whom should I solicit?	CART, logit, scoring	Predictors + observed response	Probabilities, scores, classes, targeting gains
Where should my media budget go?	Response models, MMM	Sales and spending over time	Contribution and ROI by channel

The course's common thread

1. Start from a **managerial problem** with a tension.
2. Choose **the** right method (one, two at most).
3. Collect clean data (right questions, right scales, right respondents).
4. Interpret critically: models are neither true nor decision-makers.
5. End with a **concrete, quantified recommendation**, and say what you would not do.

Numbers to know

Number	Context
$3.7–3.9 trillion	Global marketing investment per year
−40% / 34 → 37%	Heinz: promotional spending / market share
3% → 38%	Melijoe: customers / sales
72% (29/40)	Hit rate in the Enginius example
21	Minimum profiles for 5 attributes with 5 levels
85% · 60% · 90%	Use of conjoint in business
8.0 vs 5.9	Perceived network quality SFR vs Débitel
$35	Value-in-use of the O-ring
€20 → €24	Ice-cream seller's profit with discrimination
80% (246/306)	Logit correctly classified, elective
20% = 76% / 86%	Donors = donations / net margin
18.4%	Initial fundraising ratio